UGANDA CHRISTIAN UNIVERSITY
A Centre of Excellence in the Heart of Africa

FACULTY OF ENGINEERING, DESIGN AND TECHNOLOGY
DEPARTMENT OF COMPUTING AND TECHNOLOGY
EASTER 2026 SEMESTER **EXAMINATION**

PROGRAM: *BACHELOR OF SCIENCE IN DATA SCIENCE & ANALYTICS*

YEAR: 3 SEMESTER: 2

COURSE CODE: *DSC3219*

COURSE NAME: *CLOUD AND DISTRIBUTED COMPUTING*

EXAMINATION TYPE: *100% PROJECT-BASED EXAM*

PROJECT DURATION: APRIL 2026

TIME ALLOWED: *TWO WEEKS*

Examination Instructions

1. The general Uganda Christian University examination guidelines and academic & financial policies apply to this examination. Violating any of the policies by the student automatically makes this examination attempt void, even if you have completed and submitted the answer booklet.
2. This exam consists of a project to be executed in Two weeks.
    a. Assessment of the project shall be based on five milestones, evaluated during the duration of the project. Each milestone shall be evaluated out of 20 marks.
    b. At the end of the project, the following SHALL be submitted on Moodle.
        i. A well-written project (Font: Trebuchet MS, 12Pts, 1.5 spacing, justified aligned), IEEE Referencing style.
3. Every student has a responsibility to prove their contribution towards every milestone, and marks may be awarded to every student individually.

PART A:
PROJECT DESCRIPTION Distributed Data Processing Pipeline

This project is intended to leverage distributed computing to process large datasets. Using the skill learned, demonstrate the application of MapReduce model, load balancing, cluster, Virtualisation management etc. The project should be structured as follows:

a. Input Stage: In this stage, data should be stored in a distributed system (e.g., HDFS or AWS S3).
b. Processing Stage: It should be hosted on a server like Apache Spark or Hadoop cluster in the cloud to process data in parallel.
c. Result Store: The aggregated data loaded into a data warehouse (e.g., Amazon Redshift, Google BigQuery, Snowflake, and Azure Synapse Analytics).

The completed project should be documented with:

- System Architecture Diagram showing interactions between front-end, load balancers, servers, and storage.
- Scalability Analysis showing how the system handles increasing loads.
- Fault Tolerance Mechanisms showing steps taken if a node or service fails.
- Security Configuration showing the IAM policies and network security

PART B: Project-based assessment guidelines

S/N	Milestone Description	Maximum Marks
1	**MILESTONE ONE(Requirements specification- Planning)** Develop system specifications to specify: • The system functional (features) and non functional (quality attributes like security, usability, and reliability) requirements. • Defines what the system must do and its performance constraints, producing a Software Requirements Specification (SRS) document on project goals, scope, and validation criteria. Full marks awarded for: Clear software requirements, system constraints and goals of the system.	**20 %**
2	**MILESTONE TWO(System Design)** Design a cloud-based system using appropriate middleware. Full marks awarded for: Clear and documented design of the system.	**20 %**
3	**MILESTONE THREE(Secure Implementation)** Implement the system using cloud SDKs/APIs (e.g., AWS S3, Firebase). Ensure Authentication/Security of the system by implementing user authorization, authentication and securing access to files Full marks awarded for: Clear and secure system implementation with necessary	**20 %**

© Uganda Christian University | Exams for Easter 2026 Semester.

	functionalities and services.	
4	**MILESTONE FOUR(Presentation & Defense)** Present the solution with technical walkthrough including: • Live demonstration or recorded walkthrough of the solution (2-3 minutes) • Defense of asset optimization decisions and rendering choices • system implementation discussion with visual evidence • Q&A session defending design and technical decisions • Professional presentation delivery (can be recorded) Full marks awarded for: Clear articulate presentation, confident technical answers, well-justified decisions, professional delivery, evidence of deep understanding.	**20 %**
5	**MILESTONE FIVE(Technical Documentation & Report)** Submit comprehensive IEEE-formatted report (Font: Trebuchet MS, 12pt, 1.5 spacing, justified) containing: • Executive Summary: Project overview, key achievements, technical highlights (1 page) • Asset Inventory: List all models with vertex counts, origin positions, and transformation data • Detailed explanation of implementation with diagrams/screenshots • Technical justification of software choice and settings • Conclusion: Learning outcomes and challenges overcome • IEEE References: All asset sources and technical references Full marks awarded for: Professional formatting, complete documentation, clear explanations, proper IEEE references.	**20 %**
	TOTAL MARKS	**100 %**

~END OF EXAM GUIDELINES~

© Uganda Christian University | Exams for Easter 2026 Semester.